Road and mobility network	National highways, major roads and transport corridors
Demographic and economic activity	Population, urbanisation and commercial activity datasets
Electricity infrastructure	Publicly available grid and utility information
Supporting contextual information	Tourism, logistics, land-use and urban development data

Where live or detailed datasets are unavailable, teams may use publicly available, open-source, sample, or realistic synthetic datasets. All data sources and assumptions must be clearly documented.

Expected Deliverables ROUND 2: build & demo - NOT due now

Teams selecting this challenge should submit:

- **Functional prototype or web-based platform** presenting a unified view of EV charging infrastructure.
- **Interactive national GIS map** showing existing chargers, infrastructure coverage, and identified gaps.
- **Analytical methodology document** explaining the approach used for integration, gap assessment, and estimation.
- **Charger requirement estimation and recommendation module** for selected states, districts, cities, or transport corridors.
- **Decision-support dashboard** presenting relevant indicators and planning insights.
- **Presentation or demonstration video** showcasing the solution for at least one representative geography.
- **Scalability roadmap** outlining how the solution could be expanded into a nationwide planning platform.

Challenge 2: Green Auto India

Creation of an India-Specific Life Cycle Assessment Database for the Automotive Sector

The environmental footprint of a vehicle extends beyond its operation and includes

Transportation & logistics	Road, rail, coastal shipping and other freight movements
Vehicle-use phase	Energy or fuel consumption, maintenance and operational parameters
End-of-life systems	Reuse, recycling, recovery and disposal pathways
Environmental indicators	Greenhouse gas emissions and other relevant impact categories

Teams are not expected to create an exhaustive national database during the competition. They may demonstrate the proposed framework using a defined set of materials, components, processes, or vehicle systems. All sources, boundaries, assumptions, limitations, and data-quality considerations must be clearly stated.

Expected Deliverables

Teams selecting this challenge should submit:

- **Prototype India-specific automotive LCA database** covering a clearly defined scope.
- **Database architecture and data-collection framework** demonstrating how additional datasets can be incorporated.
- **Environmental impact calculation engine** for selected automotive materials, components, or vehicle technologies.
- **Web-based dashboard or visualisation** platform presenting lifecycle results and comparisons.
- **Methodology document** explaining system boundaries, assumptions, data sources, calculations, and limitations.
- **Demonstration case study** applying the proposed database to at least one automotive product, component, material, or technology.
- **Scalability roadmap** outlining how the solution could support wider adoption by ICAT, industry, researchers, and policymakers.

Competition Format

ROUND 1 starts here

Round 1: Concept & Roadmap Submission

Round 1 (submit by 30 Oct) Round 2 / Grand Finale PS page 6

ROUND 1: ONLY these are submitted now (by 30 Oct)

The submission should be concise and include:

- **Team Information:** Team Name, Team ID, Institution, and Contact Details.
- **Methodology Document:** Explaining the team's understanding of the problem and the analytical approach they intend to use (e.g., for integration, gap assessment, and estimation under Challenge 1, or the LCA framework under Challenge 2).
- **Implementation Roadmap:** A complete, step-by-step plan explaining exactly how the team intends to build their proposed solution — including the data sources to be used, tools/technologies planned, development stages, and how they intend to arrive at the final prototype ahead of the Grand Finale.

Submission Guidelines: Online submissions for this competition must be made through the official Techfest website prior to the specified deadline. Only the Team Leader is authorized to submit on behalf of the team. Submissions made by any other team member will not be considered valid, multiple submissions from the same team will not be accepted and may lead to disqualification from the competition.

Shortlisting: There is no fixed number of teams to be shortlisted. The top-performing teams, as determined by the evaluation panel, will advance to Round 2: the Grand Finale at Techfest, IIT Bombay.

ROUND 1: how the submission is scored

Evaluation Criteria for Round 1

Evaluation Criterion	Weight
Problem Understanding & Relevance	25%
Feasibility & Technical Soundness of Approach	25%
Innovation / Novelty of Proposed Solution	20%
Clarity & Completeness of Implementation Roadmap	20%
Planned Data Strategy (sources & integration approach)	10%

Total	100%

ROUND 2 (GRAND FINALE): live demo of the Expected Deliverables

Round 2: Grand Finale - Live Demonstration & Presentation (Techfest, IIT Bombay)

Shortlisted teams are expected to build their proposed solution in line with the Implementation Roadmap submitted in Round 1, and present a **live demonstration of their working prototype** before an expert jury.

The presentation should include:

- A live demonstration of the working prototype/solution, covering the Expected Deliverables listed under the chosen challenge track (see Expected Deliverables).
- A brief walkthrough of how the final solution builds on or has evolved from the roadmap submitted in Round 1.

Q&A: Following the presentation, each team will go through a Q&A / jury interaction session based on a live demonstration presented.

ROUND 2: how the finale is scored

Evaluation Criteria for Round 2

Evaluation Criterion	Weight
Technical & Analytical Robustness	25%
Data Quality & Integration	20%
Relevance & Practical Usability	20%
Innovation	15%
Scalability & Replicability	10%
Presentation & Communication	10%
Total	100%

Suggested Solution Approach (Indicative)

Round 1 Submission deadline	30th October 2026
Round 1 Result announcement	7th November 2026
Round 2 : Finale at Techfest, IIT Bombay	16th – 18th December 2026

General Rules

1. Teams must register online at techfest.org, receiving a Team ID for future reference.
2. The organizers' and judges' decisions are final and binding. Techfest reserves the right to verify participant details.
3. All submissions must be the original work of the participating team.
4. Open-source tools, libraries, models, and public datasets may be used with appropriate attribution.
5. Any synthetic data used in the solution must be clearly identified and documented.
6. Teams must disclose the use of AI tools, third-party software, external datasets, and pre-built frameworks.
7. Finalists must be available to present and defend their solution during the Grand Finale at IIT Bombay.
8. Techfest, IIT Bombay, is not responsible for late or lost entries.
9. Solutions must be original and not protected by patents, copyrights, or publications.
10. The latest information will be available on techfest.org, with registered participants notified of changes via email.
11. All technical, procedural, and competition-related queries must be submitted directly through the Techfest Competition Portal under the Doubt section. Participants are strictly requested to regularly check the portal for answers and updates
12. Official communication for important announcements will be through the email siddhi@techfest.org. Queries submitted via email will not be entertained

Certificate Policy and Recognition

- Certificates of merit and national recognition for finalists during